

# Monthly Intelligence Report

Edition 002 — The Tyrol-Vorarlberg Alpine Corridor

Where the energy transition is outrunning the grid. edition · May 2026 · SSI v4.0.2

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## SECTION A

### Executive Summary

#### SUBSTATIONS SCORED

1,406

1,144 HV · 262 MV

#### GRID LINES MAPPED

11,267

~15,000 km coverage

#### MC SIMULATIONS

14.06M

10,000 per substation

#### PUBLIC DATA SOURCES

35

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 346,000+ source data points per run, executes 14.06 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and produces full uncertainty quantification across 1,406 substations spanning 85 Bezirke (administrative districts) and 11,267 grid lines covering ~15,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 35 verified public sources, making the methodology fully auditable and reproducible. The Index captures Austria's unique challenges: Alpine infrastructure exposure, hydropower dependency (60%+ of renewable generation), east-west transmission constraints, and cross-border interconnection with Germany, Switzerland, Italy, and the Czech Republic. This Austrian edition prioritises the critical Tyrol-Vorarlberg Alpine Corridor, where APG transmission bottlenecks intersect with the highest hydropower concentration in Central Europe.

## Substation in Focus

### Lieserhofen

Spittal an der Drau, Kärnten · 110 kV

0.331

R\_FINAL

Medium

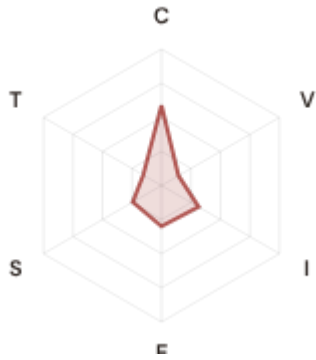
CLASSIFICATION

0.138

CI WIDTH

81th

FLEET PERCENTILE



#### COMPONENTS

**C** Continuity: 0.590 / 0.30 (197%)

**I** Infrastructure: 0.315 / 0.25 (126%)

**S** Saturation: 0.244 / 0.20 (122%)

**V** Voltage: 0.144 / 0.10 (144%)

**E** Economic: 0.300 / 0.10 (300%)

**T** Transition: 0.155 / 0.05 (309%)

#### MODIFIERS

**R3 Consequence:** 0.953 ↓ dampens

**R4 Graph Criticality:** 0.945 ↓ dampens

**R6a Restoration:** 1.041 ↑ amplifies

**R6b Network Topology:** 1.000 → neutral

**R7 Digital Readiness:** 0.987 ↓ dampens

This month's spotlight — automatically selected for May 2026 — is **Lieserhofen** in Spittal an der Drau, Kärnten. With an R\_final of 0.331 (Medium band, 81th fleet percentile), its risk profile is dominated by the T (Transition) component at 309% of its weight ceiling, followed by E (Economic) at 300%. Modifiers collectively shift R\_base by 7.2%, reflecting the substation's specific geographic, socio-economic, and network context.

#### SECTION B — RECURRING MODULE

## Cyber & Economic Exposure Monitor

**Why this section exists:** Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Austria's Energiewende transition planning and grid modernisation initiative require.

#### CYBER HIGH EXPOSURE

416

29.6% of fleet

#### BLIND SPOTS

0

High/Critical + R7 < median

#### AVG R7 MODIFIER

1.0013

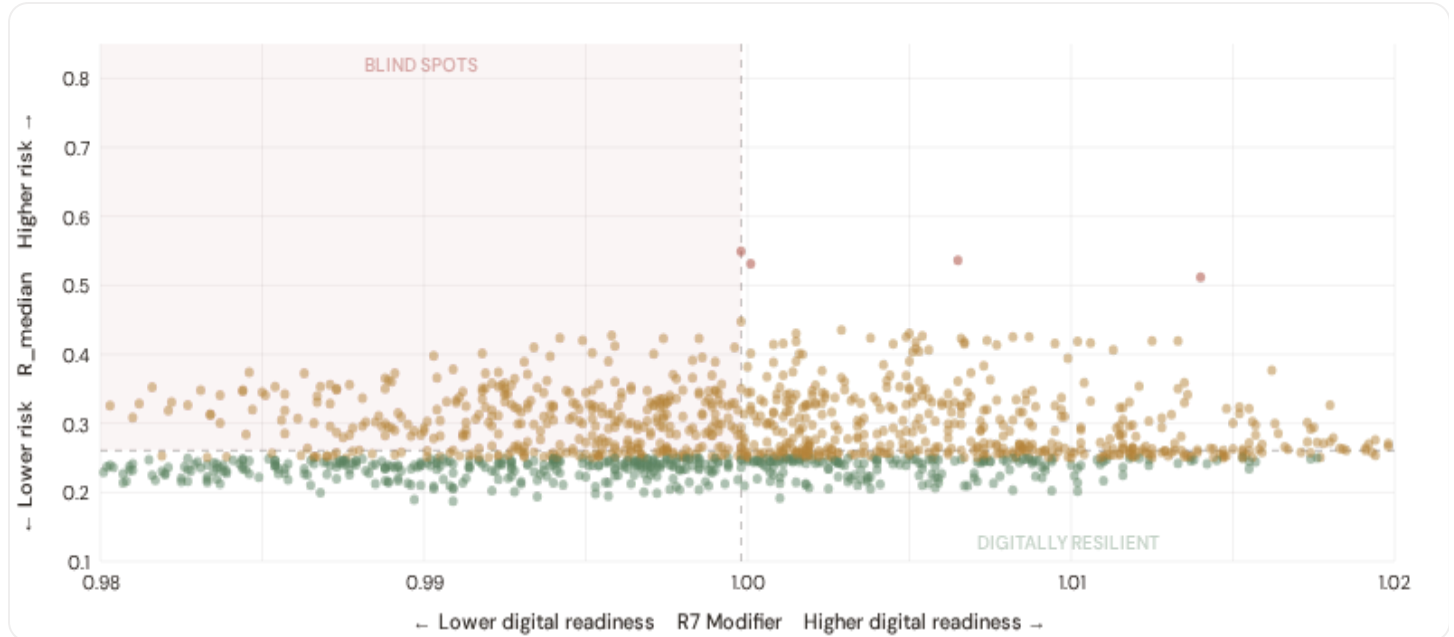
Range [0.99, 1.05]

#### HIGH-CONSEQUENCE SUBS

0

## B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R\_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **0 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ( $R7 < 0.9998$ ). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in . Across the full fleet, 416 substations (29.6%) carry a HIGH cyber-exposure classification, while 420 (29.9%) are in the LOW tier.

## B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

### Capital-Intensive / High-Consequence

Est. VoLL: Est. high VoLL

Heavy industry, critical infrastructure -- highest economic loss per interruption

**354** substations (25.2%) High/Critical: **0** (0.0%) Avg R: **0.268** Avg E: **0.303** / 0.10



### Industrial / Medium–Large Enterprise

Est. VoLL: Est. medium–high VoLL

Diversified industry, logistics, processing -- significant continuity requirements

**349** substations (24.8%) High/Critical: **2** (0.6%) Avg R: **0.283** Avg E: **0.303** / 0.10



### Commercial / SME–Dense

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

**354** substations (25.2%) High/Critical: **2** (0.6%) Avg R: **0.284** Avg E: **0.303** / 0.10



### Light / Agricultural / Remote

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

**349** substations (24.8%) High/Critical: **3** (0.9%) Avg R: **0.294** Avg E: **0.304** / 0.10

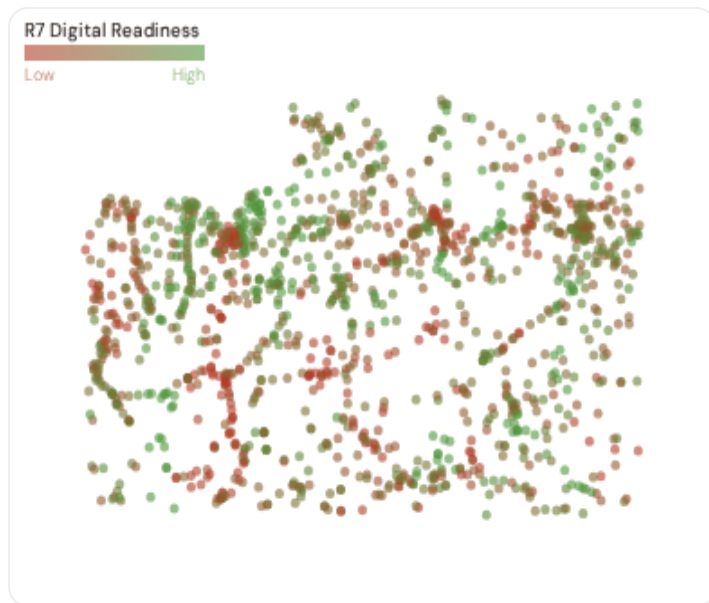


**Value of Lost Load (VoLL) context:** ACER's 2023 estimate for Austria places average VoLL at €11.2/kWh for industrial customers and €3.0/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **0 substations** combine capital-intensive economic fabric ( $R3 \geq 1.14$ ) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Tirol (105), Salzburg (48), Vorarlberg (40)** — regions where industrial corridors depend on uninterrupted power for continuous processes (paper, chemicals, food). A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. Austria's distributed economic fabric — strong across all Bundesländer — means the total economic exposure is spread more evenly than in less balanced economies. This is a structural advantage, but it does not eliminate the urgency of identifying and protecting the highest-consequence substations. This cross-referencing of economic fabric with grid condition is unique to the SSI.

## B.3 Bezirk Digital Readiness Ranking

Bezirke (districts) ranked by average R7 modifier — lower values indicate weaker digital infrastructure. Cross-referenced with average R<sub>median</sub> to identify Bezirke combining digital exposure with high grid stress.



#### WORST DIGITAL READINESS — TOP 15 BEZIRKE

Longer bar = higher R7 = better digital infrastructure.

|   |                                        |             |        |
|---|----------------------------------------|-------------|--------|
| 1 | Vorarlberg                             | <div></div> | 0.9969 |
| 2 | Salzburg                               | <div></div> | 0.9973 |
| 3 | Wien                                   | <div></div> | 0.9992 |
| 4 | Burgenland <span>▲ high R</span>       | <div></div> | 1.0002 |
| 5 | Oberösterreich <span>▲ high R</span>   | <div></div> | 1.0006 |
| 6 | Kärnten <span>▲ high R</span>          | <div></div> | 1.0007 |
| 7 | Tirol                                  | <div></div> | 1.0026 |
| 8 | Niederösterreich <span>▲ high R</span> | <div></div> | 1.0031 |
| 9 | Steiermark <span>▲ high R</span>       | <div></div> | 1.0047 |

Regional analysis reveals digital infrastructure variation across Austria's 9 Bundesländer. **Vorarlberg** has the lowest average R7 (0.9969), while **Steiermark** leads at 1.0047 — a spread of 0.0079. **9 Bundesländer** combine below-median digital readiness with above-median grid risk, making them priority targets for E-Control-coordinated modernisation investment.

## B.4 Monthly Pulse

**Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor across the Austrian fleet. Key reference points: fleet average R7 = 1.0013, median = 0.9998; 416 substations classified HIGH cyber-exposure; 0 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging Bundesländer where digitalisation investments translate into measurable R7 shifts.

### SECTION C

## Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. Austrian sources include: E-Control monitoring reports, Statistik Austria STATPOP, APG Netzentwicklungsplan, ZAMG/GeoSphere climate data, and BEV geographic data. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet.

TOTAL SOURCES

**35**

95 variables

WEEKLY / MONTHLY

**5**

High-frequency feeds

QUARTERLY / ANNUAL

**22**

Regular refresh cycle

STATIC / DERIVED

**8**

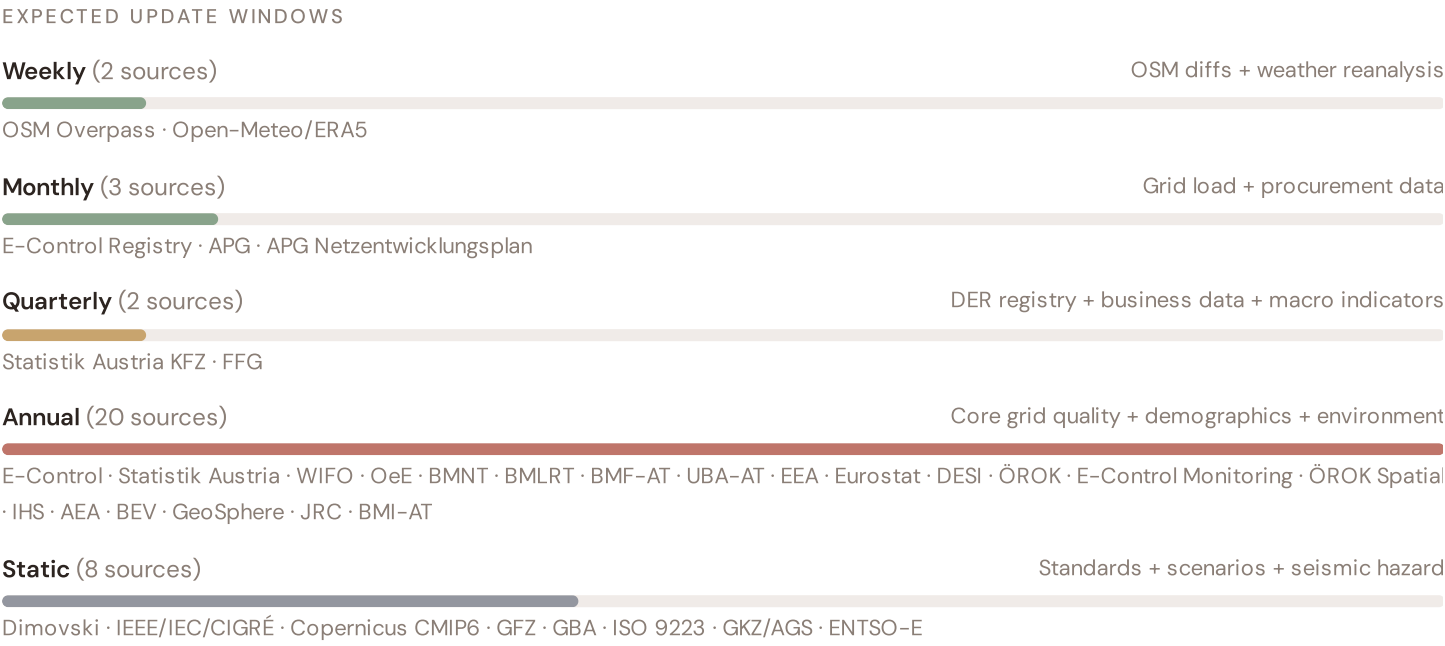


## Data Source Registry — May 2026

|          |                                    |            |                  |        |
|----------|------------------------------------|------------|------------------|--------|
| OSM      | OSM Power Infrastructure           | WEEKLY     | Node/edge        | 3 vars |
| APG      | APG Transmission System Operator   | MONTHLY    | Control area     | 3 vars |
| APG-N    | APG Netzentwicklungsplan           | MONTHLY    | TSO zone         | 2 vars |
| BMI-AT   | BMI Civil Protection Austria       | CONTINUOUS | Bezirk           | 1 var  |
| STAT-KFZ | Statistik Austria KFZ Registry     | QUARTERLY  | Bezirk           | 1 var  |
| FFG      | FFG Innovation Agency Austria      | QUARTERLY  | Bezirk           | 1 var  |
| ECONT    | E-Control Austria                  | ANNUAL     | Bezirk           | 8 vars |
| ECREG    | E-Control Grid Registry            | ANNUAL     | Bezirk           | 5 vars |
| STAT     | Statistik Austria                  | ANNUAL     | Bezirk (GKZ)     | 8 vars |
| UBA-AT   | Umweltbundesamt Austria            | ANNUAL     | Bezirk / station | 4 vars |
| JRC      | JRC DSO Observatory                | ANNUAL     | DSO-level        | 2 vars |
| EEA      | EEA Air Quality e-Reporting        | ANNUAL     | ~1 km + station  | 3 vars |
| EURO     | Eurostat Energy Statistics         | ANNUAL     | NUTS2            | 3 vars |
| DESI     | DESI Digital Economy Index         | ANNUAL     | EU Regional      | 2 vars |
| BMNT     | BMNT Sustainability & Tourism      | ANNUAL     | Bundesland       | 2 vars |
| BMLRT    | BMLRT Regional Development         | ANNUAL     | Bundesland       | 1 var  |
| BMF-AT   | BMF Austria Fiscal Statistics      | ANNUAL     | Bezirk           | 1 var  |
| BEV      | BEV Federal Office of Metrology    | ANNUAL     | Bundesland       | 1 var  |
| WIFO     | WIFO Economic Research             | ANNUAL     | Regional         | 2 vars |
| OEE      | Oesterreichs Energie (Association) | ANNUAL     | Bundesland       | 2 vars |
| OEROK    | ÖROK Regional Indicators           | ANNUAL     | Bezirk           | 2 vars |
| ECMON    | E-Control Monitoring Report        | ANNUAL     | DSO-level        | 2 vars |
| OROK-S   | ÖROK Spatial Planning              | ANNUAL     | Bezirk           | 1 var  |
| IHS      | IHS Institute for Advanced Studies | ANNUAL     | Bundesland       | 1 var  |
| AEA      | Austrian Energy Agency             | ANNUAL     | Bundesland       | 1 var  |
| GBA      | GBA Geological Survey Austria      | STATIC     | Bezirk           | 3 vars |

|          |                                   |         |                  |         |
|----------|-----------------------------------|---------|------------------|---------|
| D10      | Copernicus CDS / ERA5             | STATIC  | 0.25° (~25 km)   | 4 vars  |
| DIM      | Dimovski et al. (2025)            | STATIC  | Municipal        | 3 vars  |
| IEEE-1   | IEEE C57.91 / IEC 60076           | STATIC  | Asset-level      | 16 vars |
| GFZ      | GFZ German Research Centre        | STATIC  | ~5 km            | 2 vars  |
| GKZ      | Gemeindekennziffer / GKZ Registry | STATIC  | Bezirk (5-digit) | 1 var   |
| ISO-9223 | ISO 9223 Corrosion                | DERIVED | Bezirk           | 1 var   |
| ENTSE    | ENTSO-E Transparency              | HOURLY  | Bidding zone     | 2 vars  |
| ZAMG     | GeoSphere Austria (ex-ZAMG)       | DAILY   | ~1 km            | 3 vars  |
| D-0M     | Open-Meteo / ERA5                 | HOURLY  | ~1 km            | 1 var   |

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot





BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **1,406 substation scores remain unchanged** this edition. The fleet median R of 0.261 (mean 0.282) reflects a distribution skewed toward the lower-to-medium bands, with 34.7% in Low and 64.8% in Medium. The first material score changes are expected when E-Control publishes the updated TIQE tables (affecting C1–C4 and R6a) and when APG refreshes the DER registry (affecting T1).

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across PO, P1, and P2 releases

|    |          |                                                                                     |           |
|----|----------|-------------------------------------------------------------------------------------|-----------|
| F1 | NEW      | New T component — Energy Transition Exposure (T1)                                   | \$2, \$4  |
| F2 | NEW      | New R6a — Restoration Speed Modifier (CAIDI-based)                                  | \$5       |
| F3 | ENHANCED | R3 enhanced — Energy Poverty Vulnerability (V_socio)                                | \$5       |
| F4 | ENHANCED | R4 enhanced — Graph-theoretic betweenness + bridge detection                        | \$5       |
| F5 | ENHANCED | R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)                                   | \$5       |
| F6 | ENHANCED | R7 Digital Readiness — Bezirk-level DESI model                                      | \$5       |
| L1 | NEW      | New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo                          | \$2, \$8  |
| L2 | ENHANCED | E2 Innovation Enrichment — HRST + startup density                                   | \$6       |
| L3 | ENHANCED | V_socio Fiscal Enrichment — BMF + ÖROK + energy price                               | \$5       |
| L4 | ENHANCED | R3 Demographic Shift Amplifier — Statistik Austria net migration                    | \$5       |
| L5 | DATA     | 95/95 variables operational (100%). 35 data sources total.                          | \$8, \$12 |
| G1 | DATA     | E-Control Registry upgraded to bulk download — Bezirk-level DER registry            | \$12      |
| G2 | DATA     | GBA upgraded to WMS live API — Bezirk-level geological data                         | \$12      |
| G3 | DATA     | OSM upgraded to Overpass API — 1,406 real substations                               | \$12      |
| G4 | NEW      | R6b Network Topology modifier — centrality + ring analysis from OSM graph           | \$5       |
| G5 | NEW      | Network & Topology layer (H): 7 variables — APG Netzentwicklungsplan, ring analysis | \$12      |
| G6 | NEW      | Output Scores layer (I): 7 variables — risk_score, ETTc, stationary probs           | \$12      |

SECTION D — MONTHLY DEEP-DIVE

The Tyrol-Vorarlberg Alpine Corridor: Transmission Bottlenecks Meet Hydropower Density

**Deep-dive rotation:** Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Tyrol-Vorarlberg corridor · **002** Oberösterreich (Industrial grid, VOEST) · **003** Wien (Urban density, Wiener Netze) · **004** Niederösterreich (Wind integration, largest Bundesland) · **005** Steiermark (Green hydrogen, industrial transition) · **006** Kärnten (Southern Alpine, cross-border Italy) · **007** Salzburg (Hydropower hub, tourism load) · **008** Burgenland (Wind champion, Pannonian corridor) · **009** Vorarlberg (Isolated west, CH/DE cross-border) · **010** Tirol (Brenner transit, ski tourism seasonality) · **011** Annual Review · **012** Methodology Deep-Dive.

## D.1 Why the Alpine Corridor, Why Now

CORRIDOR SUBSTATIONS

**593**

126 HV · 467 MV

HIGH BAND

**0**

0.0% of corridor

CORRIDOR MEDIAN R

**0.248**

Fleet median: 0.261

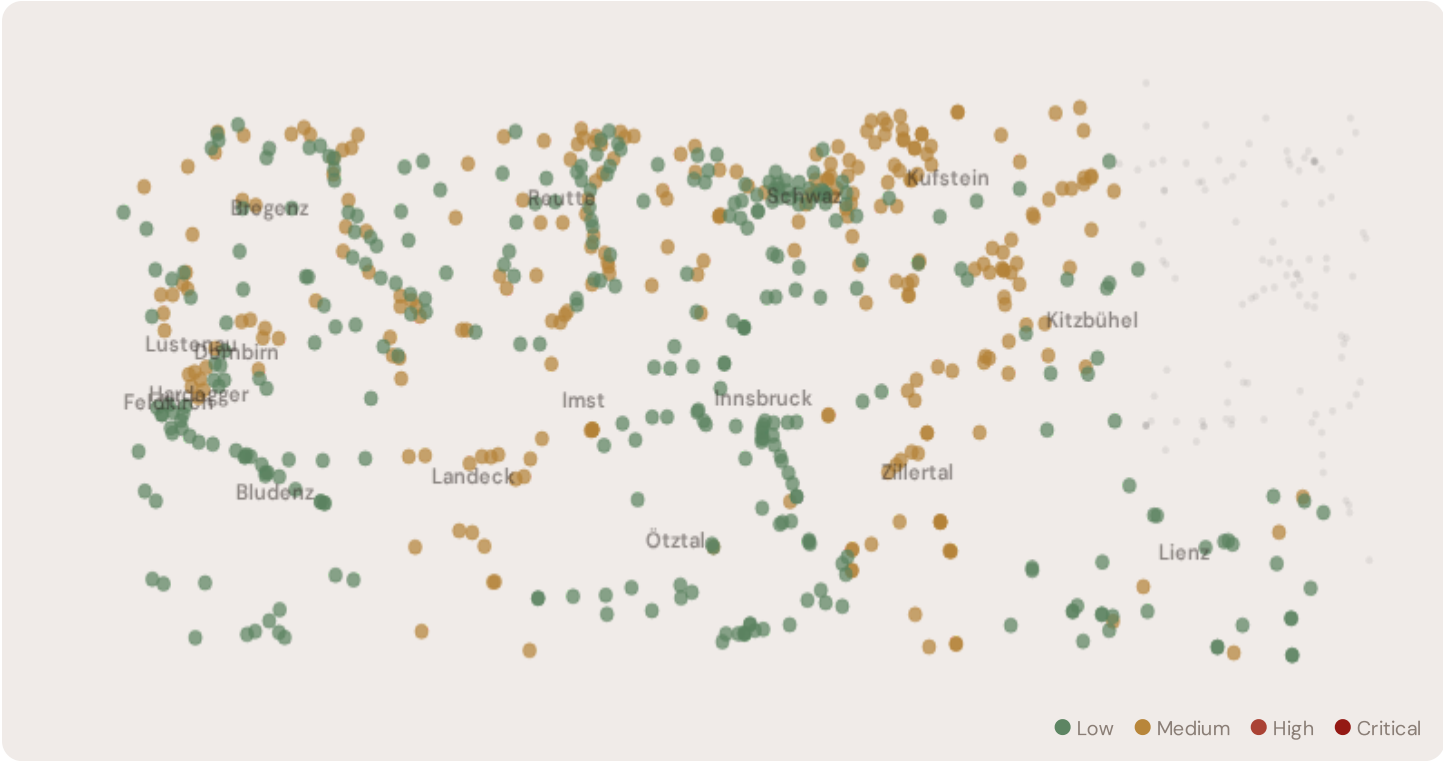
WORST PROVINCE

**Landeck**

Avg R: 0.338 · 18 substations

The Tyrol-Vorarlberg Alpine Corridor hosts **593 substations**, concentrated in Austria's two westernmost Bundesländer. This region presents a unique combination of engineering and economic challenges: Alpine terrain creates infrastructure vulnerability, hydropower generates 60%+ of Austria's renewable supply but introduces seasonal volatility, and APG's transmission network faces critical bottlenecks (380 kV Salzburgleitung, Arlberg) limiting east-west power flow and forcing reliance on transit through Germany and Switzerland. Risk profile is concentrated: **0 substations (0.0%)** are classified High, with only **324** in the Low band. 19% of corridor substations score above the national fleet median of 0.261. The corridor median R of 0.248 places Tirol-Vorarlberg among Austria's most stressed transmission corridors.

## D.2 Corridor Map and Risk Scoring



| SUBSTATION                | PROVINCE  | R_FINAL | BAND   | C     | V     | I     | E     | S     | T     | ALERT |
|---------------------------|-----------|---------|--------|-------|-------|-------|-------|-------|-------|-------|
| Malga Mare                | Landeck   | 0.356   | Medium | 0.257 | 0.145 | 0.314 | 0.292 | 0.252 | 0.147 | —     |
| Zillertal Junction-108    | Zillertal | 0.355   | Medium | 0.268 | 0.158 | 0.350 | 0.329 | 0.242 | 0.149 | —     |
| Umspannwerk Nauders       | Landeck   | 0.353   | Medium | 0.264 | 0.148 | 0.322 | 0.309 | 0.259 | 0.148 | —     |
| Glorenza                  | Landeck   | 0.352   | Medium | 0.256 | 0.157 | 0.323 | 0.313 | 0.264 | 0.148 | —     |
| Tiwag Umspannwerk Ötztal  | Imst      | 0.348   | Medium | 0.260 | 0.154 | 0.327 | 0.299 | 0.247 | 0.146 | —     |
| APG Umspannwerk Westtirol | Imst      | 0.348   | Medium | 0.251 | 0.157 | 0.335 | 0.295 | 0.252 | 0.156 | —     |
| Landeck Junction-125      | Landeck   | 0.348   | Medium | 0.250 | 0.157 | 0.329 | 0.311 | 0.241 | 0.144 | —     |
| Lappach – Lappago         | Zillertal | 0.347   | Medium | 0.261 | 0.151 | 0.333 | 0.313 | 0.268 | 0.155 | —     |
| UW Tobadill               | Landeck   | 0.347   | Medium | 0.241 | 0.151 | 0.312 | 0.300 | 0.265 | 0.145 | —     |
| Zillertal Junction-124    | Zillertal | 0.346   | Medium | 0.272 | 0.148 | 0.345 | 0.308 | 0.253 | 0.135 | —     |

... 583 additional substations — [explore full corridor in Digital Twin](#) →

### D.3 Component Analysis

COMPONENT AVERAGES — ALPINE VS FLEET

|                    |                       |
|--------------------|-----------------------|
| C — Continuity     | 0.221 vs 0.331 (−33%) |
| I — Infrastructure | 0.320 vs 0.320 (0%)   |
| S — Saturation     | 0.257 vs 0.257 (0%)   |
| V — Voltage        | 0.150 vs 0.150 (−0%)  |
| E — Economic       | 0.303 vs 0.303 (−0%)  |
| T — Transition     | 0.150 vs 0.150 (−0%)  |

MODIFIER AVERAGES — ALPINE VS FLEET

|                  |       |               |
|------------------|-------|---------------|
| R3 Consequence   | 0.954 | fleet 0.957 ↓ |
| R4 Graph Crit.   | 1.021 | fleet 1.010 ↑ |
| R6a Restoration  | 1.039 | fleet 1.032 ↑ |
| R6b Seismic      | 1.000 | fleet 1.000 → |
| R7 Digital Read. | 1.001 | fleet 1.001 → |

The dominant risk driver across the Alpine Corridor is **E (Economic)**, averaging **0.303** against a fleet average of 0.303 — occupying **303% of its weight ceiling**. The second contributor is **T (Transition)** at **0.150** vs fleet 0.150. Among modifiers, the **R6b seismic modifier** averages **1.000**, capturing the Alpine region's moderate seismic exposure — a unique risk dimension captured by the SSI. The **Transition component (T)** at **0.150** reflects the accelerating DER penetration and renewable integration driving portfolio stress in these corridors.

### D.4 Province Breakdown

PROVINCE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

|           |                                     |
|-----------|-------------------------------------|
| Landeck   | 18 subs · avg R = 0.338 · H:O M:18  |
| Imst      | 6 subs · avg R = 0.335 · H:O M:6    |
| Zillertal | 30 subs · avg R = 0.332 · H:O M:30  |
| Lustenau  | 7 subs · avg R = 0.268 · H:O M:7    |
| Kufstein  | 103 subs · avg R = 0.258 · H:O M:88 |
| Reutte    | 116 subs · avg R = 0.250 · H:O M:61 |
| Bregenz   | 77 subs · avg R = 0.249 · H:O M:36  |
| Schwaz    | 49 subs · avg R = 0.245 · H:O M:14  |
| Kitzbühel | 9 subs · avg R = 0.243 · H:O M:1    |
| Lienz     | 37 subs · avg R = 0.241 · H:O M:5   |
| Ötztal    | 29 subs · avg R = 0.237 · H:O M:1   |
| Innsbruck | 59 subs · avg R = 0.236 · H:O M:2   |



The province-level breakdown reveals concentration of risk. **Landeck** leads with a mean R of 0.338 across 18 substations, while **Dornbirn** scores 0.195 — a spread of 0.143. APG's transmission network focuses on interconnection and east-west power transfer; at this level, asset-by-asset granularity is essential for investment decisions. The spatial pattern identifies the specific bottleneck substations constraining the Brenner transit corridor and the Arlberg transmission link.

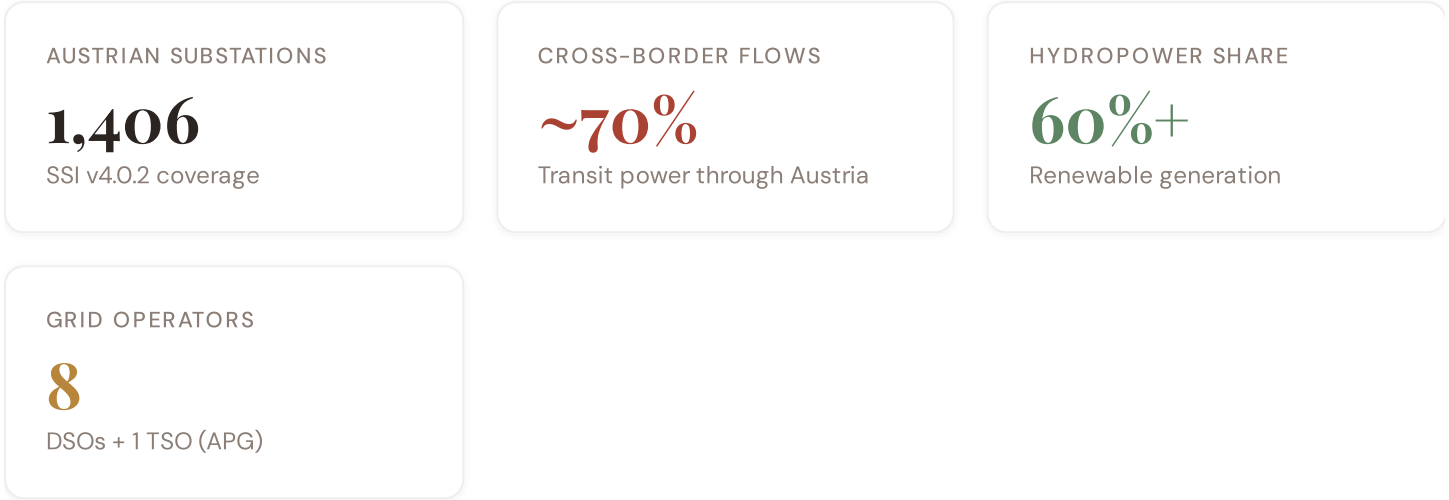
## D.5 Implications

**O Alpine Corridor substations** score  $R \geq 0.650$ , in the upper quartile of national risk. For Austrian grid modernisation priorities, this analysis identifies **Landeck** as the highest-priority target — where transmission constraints are acute, hydropower integration is intensive, and cross-border flows are heaviest. The SSI's uniqueness is that it connects engineering condition (continuity, infrastructure age) with network topology (R4) and geographic exposure (Alpine seismic, climate degradation trends). This is the evidence base for APG's €2.5B+ transmission investment programme.

### SECTION E — RECURRING MODULE

## European Context Card

**How this works:** Each edition includes one European Context Card to position Austria within the ENTSO-E and ACER benchmarking framework. This edition: **Austria as a Transit Hub** — because the Tyrol-Vorarlberg corridor analysis reveals Austria's critical role in European power flows. Future editions rotate through: Hydropower vs Nordic peers, Cross-border pricing, Renewable penetration vs. EU average, Grid digitalisation benchmarking (DESI), etc. The underlying data updates annually with ACER reporting cycles.



**Austria's position in Central Europe:** Austria sits at the intersection of four major European grid zones — German Übertragungsnetz to the north, Italian Terna to the south, Swiss Swissgrid to the west, and Czech operators to the east. APG (Austrian Power Grid) manages the 380/220/110 kV transmission network spanning 10,600 km across all 9 Bundesländer. The 2023 ACER grid adequacy assessment rated Austria as "adequate" but flagged transmission constraints in the Tyrol-Vorarlberg Alpine region as the single largest bottleneck limiting Renewable-to-load matching during high-wind periods. The SSI Index provides the substation-level intelligence that Austria's Energiewende planning requires — identifying not regional aggregate problems, but specific assets that demand urgent action.

## E.1 Austria in European Benchmarking

### RENEWABLE PENETRATION

**Austria: 76%** renewable electricity (2023) — among highest in EU. Driven by hydropower (60%) + wind (18%) + solar (4%). This creates unique grid stress: seasonal hydropower variation (dry summers) combined with wind volatility (intermittent Alpine flows) creates a transmission system under constant pressure. The ENTSO-E 2023 Winter Outlook identified Austria as a "critical node" in European power flows.

### DIGITALISATION (DESI)

**Austria: Rank 8/27 EU** on Digital Economy & Society Index (2023). E-Control's mandatory smart meter rollout (65% coverage) and APG's digitalisation programme are advancing, but rural Alpine areas lag. The SSI's R7 cyber-readiness modifier captures this gradient — identifying DSOs combining high grid stress with below-average digital infrastructure, prioritising investment targets for E-Control-funded modernisation.

## E.2 Cross-Border Interconnection

Austria is the geographic linchpin between North-South Europe (Germany ↔ Italy transit) and East-West (Germany ↔ Czech ↔ Hungary). Approximately 70% of power transmitted through the Austrian grid originates from other countries. This creates a structural vulnerability: any outage at a critical juncture (e.g., Brenner corridor, Arlberg transmission) cascades across European price zones. The SSI Index's R4 (Graph Criticality) modifier quantifies this precisely — identifying which Austrian substations are highest-impact on the European grid topology.

### SECTION F — RECURRING MODULE

## SSI Enhanced Neural Network — Monthly Topic

**From grid intelligence to BESS valuation platform.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DSO and to the community?"* Each edition explores one building block of the SSI-ENN's five-layer architecture, showing how SSI Index data feeds directly into investment-grade analytics.

#### Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

#### Layer 2 — Valuation

DNO + Community value stacks, TCO, real options

#### Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

#### Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

# Data Architecture — Component Taxonomy

*How the SSI's 6 components and 20 sub-metrics are sourced and validated*

The SSI-ENN's Layer 1 formally specifies the SSI v4.0 Index — its six components (Continuity, Voltage, Infrastructure, Economic, Saturation, Transition) and sub-metrics. Each is independently sourced from one of 35 Austrian public data sources, validated, and normalised to [0,1] scale before aggregation. This architecture is the foundation: without substation-level grid intelligence, neither the BESS valuation engine nor the neural network can operate effectively. The component taxonomy maps directly to downstream valuation — C component feeds avoided-outage calculation, E component feeds community value stack.

KEY CONCEPTS

→ 6 components, 20 sub-metrics, 95 variables

→ Min-max normalisation

→ Component weights:  $0.30 \cdot C + 0.10 \cdot V + 0.25 \cdot I + 0.10 \cdot E + 0.20 \cdot S + 0.05 \cdot T$

→ Each sub-metric traceable to public Austrian source

LIVE SSI DATA CONNECTION

The SSI Index currently scores **1,406 Austrian substations** across all 9 Bundesländer. Fleet mean R = 0.282. The foundation is 35 verified public sources feeding 95 variables.

Coming next month: 

LAYER 2

 DNO Value Stack (D1–D5) — *What a BESS is worth to the distribution network operator*

SECTION G

## Looking Ahead

#### NEXT MONTH — EDITION 003

### Tyrolean Alpine Hydro Backbone

Deep-dive into Tirol's grid risk profile — substation map, component analysis, province breakdown. European Context  
Card: undefined.

#### NEXT DATA REFRESH

### Q3 2026 — 2 Quarterly Sources

**Statistik Austria KFZ, FFG** are due for refresh in Q3. Impact: DER registry updates (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **E-Control Austria** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

#### FLEET SPOTLIGHT

### o Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **O substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 003 will be published on the second Thursday of June 2026. Deep-dive region: **Tirol**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu).

**Feedback welcome.** The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Austrian utility, E-Control, Stadtwerk, or academic institution and would like to contribute data or methodology feedback, please contact us at [ssi\\_index@ikenga.eu](mailto:ssi_index@ikenga.eu). The SSI's credibility depends on open scrutiny.

## SSI Monthly Intelligence Report — Edition 002

Published 14 May 2026 · SSI Index v4.0.2 · 1,406 substations · 11,267 grid lines · 6 components · 20 metrics · 6 modifiers  
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5